Vide Game AI - Decision Tree CRUD for NPC Behavior

Jahnavi
Priyanka

Introduction

- NPCs (Non-Player Characters) make decisions in games
- Actions include:
    - Attack
    - Patrol
    - Run
    - Search
- Decision Trees help simulate intelligent behavior

Problem Statement

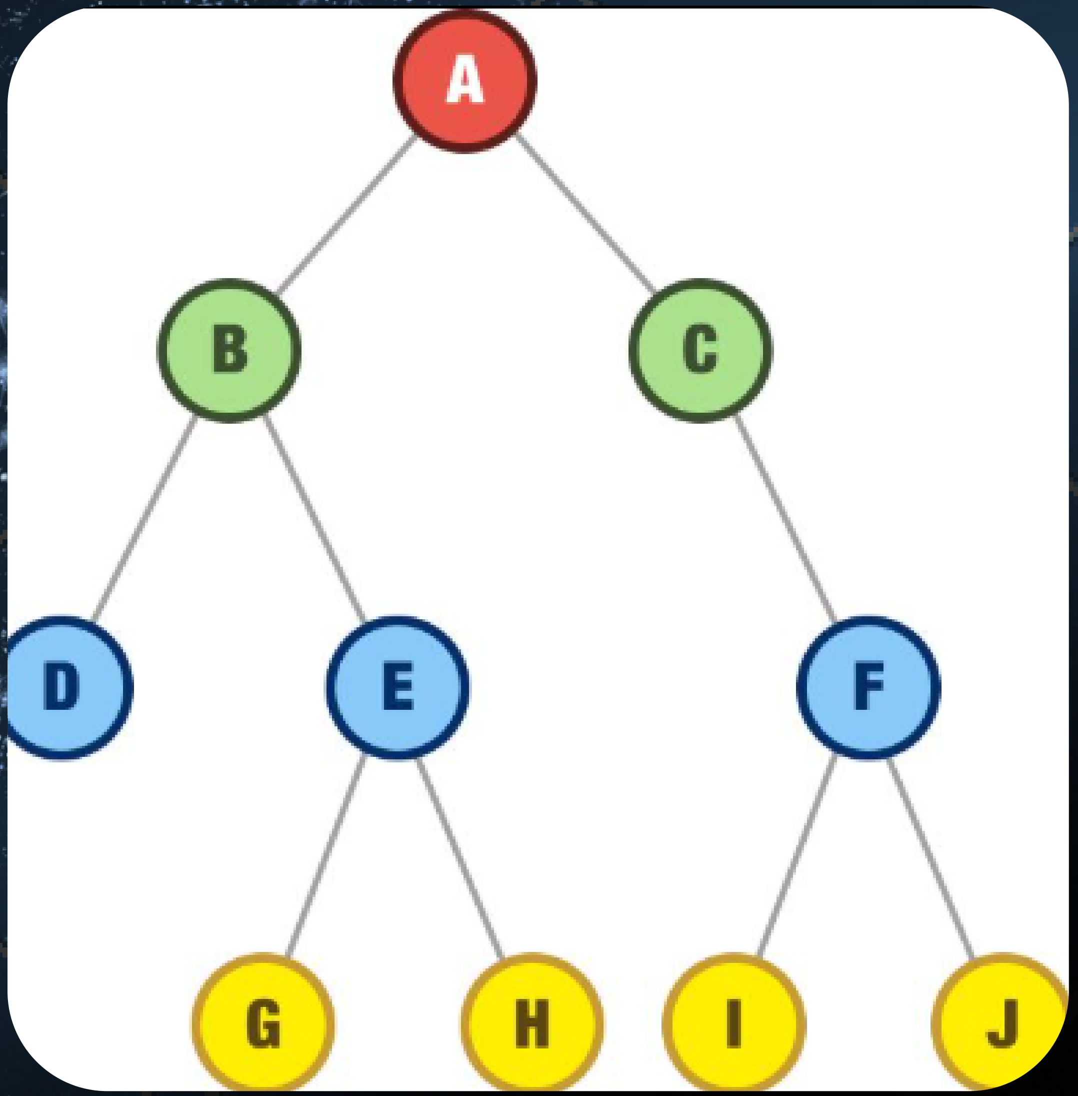
NPCs need smart decision-making

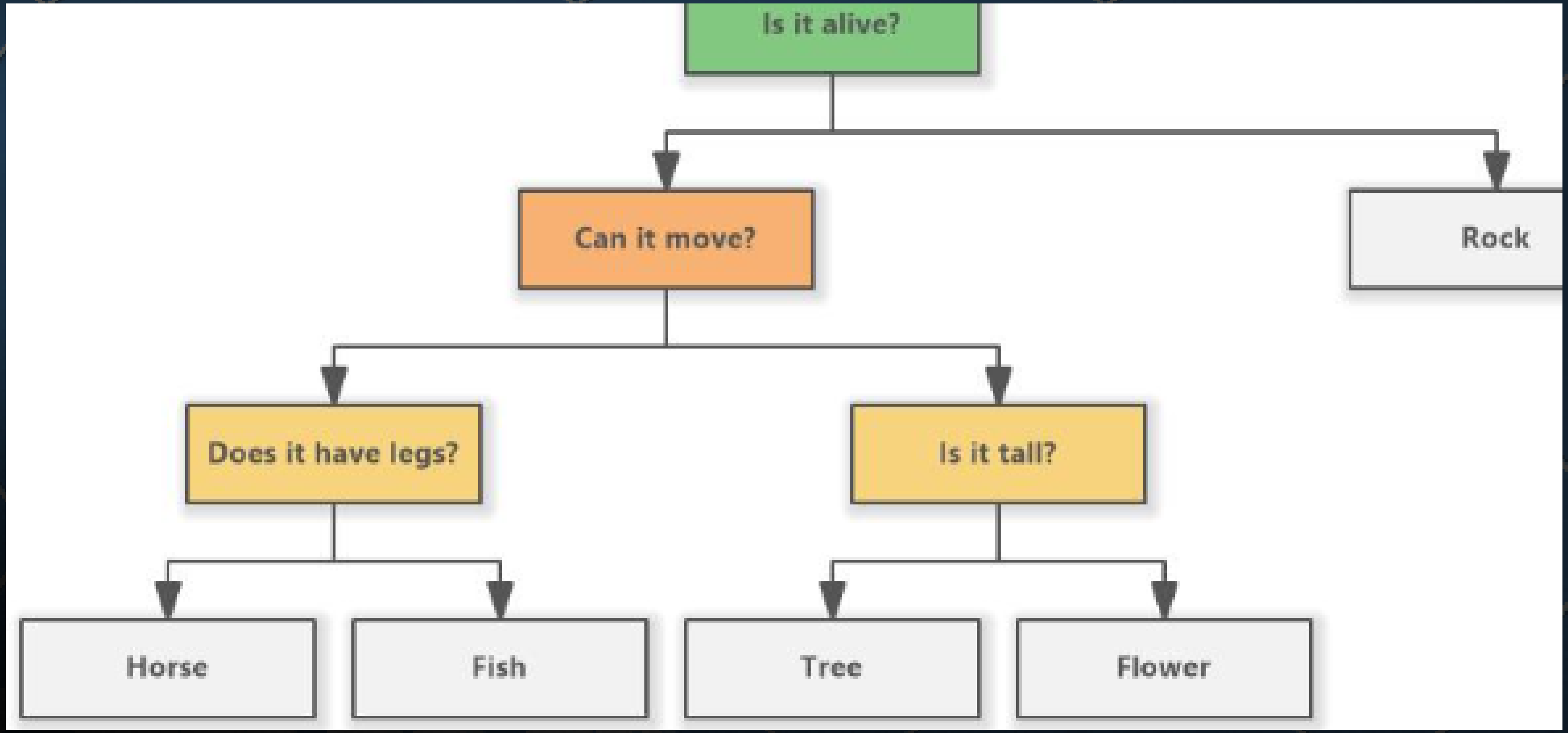
Manual logic is complex

Need system to:

- Add decisions
- Update decisions
- Delete decisions
- Search decisions

Solution: Decision Tree + CRUD Operation





Data Structure Used
Binary Tree
Node contains:
ID
Decision
Left child
Right child

Node Structure (Code)

```
struct Node
{
int id;
char decision[100];
struct Node *left;
struct Node *right;
};
```

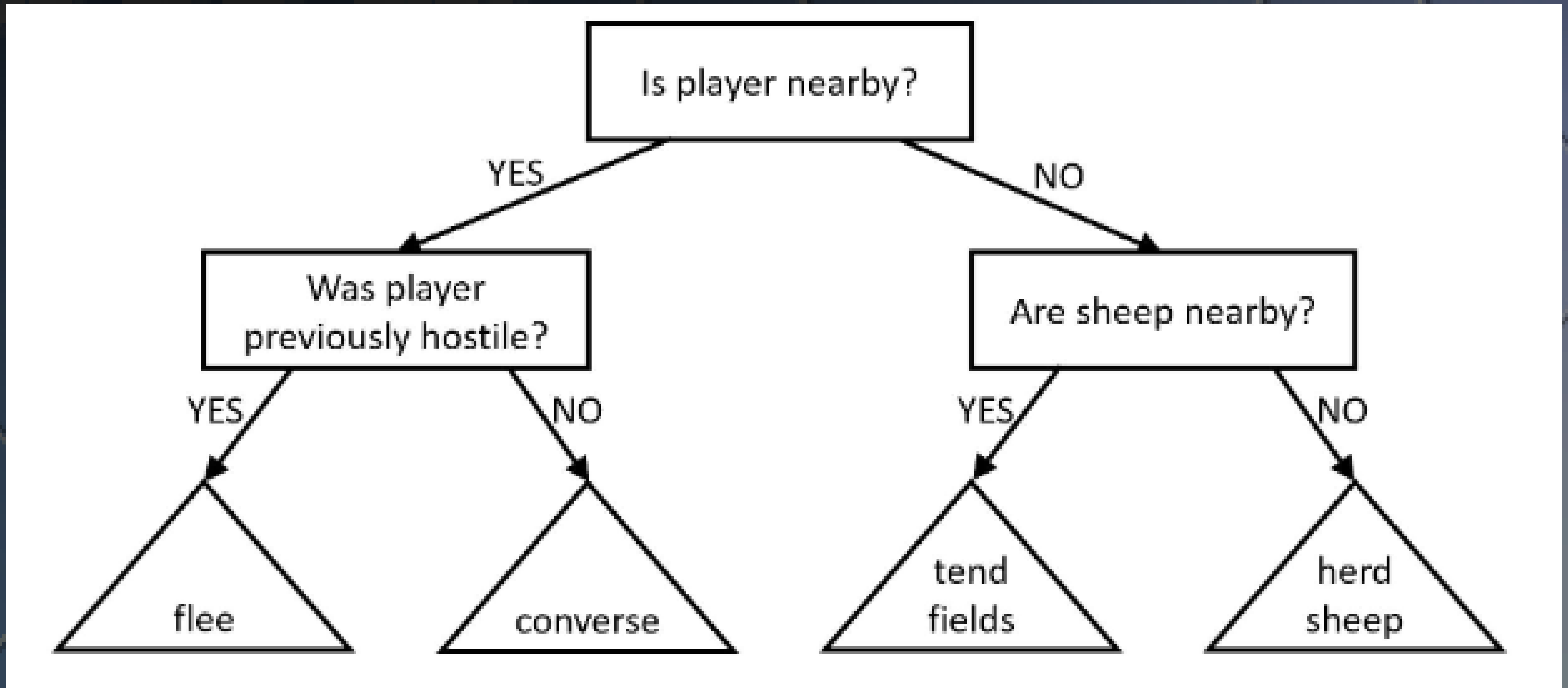
System Design

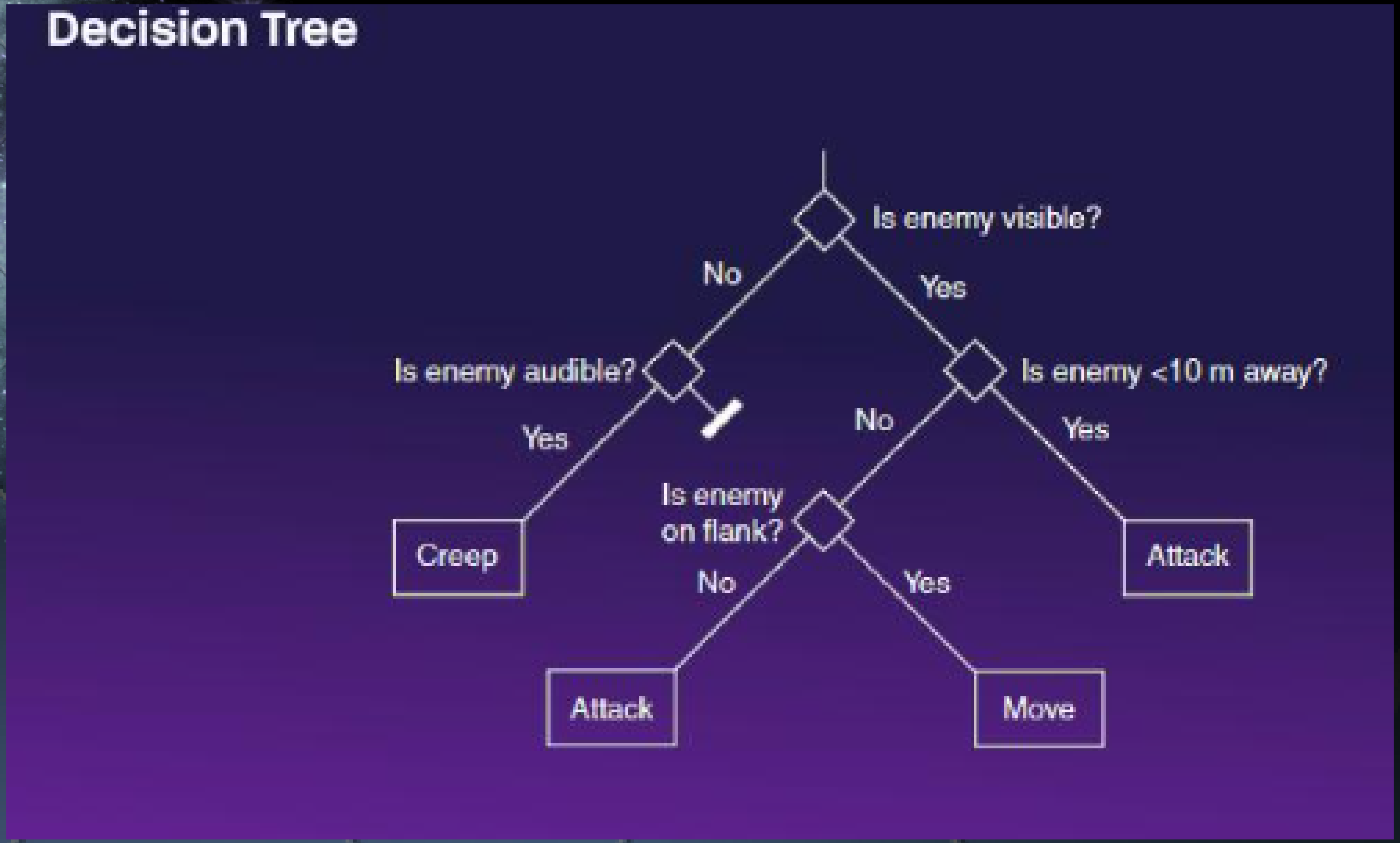
Example Decision Tree:

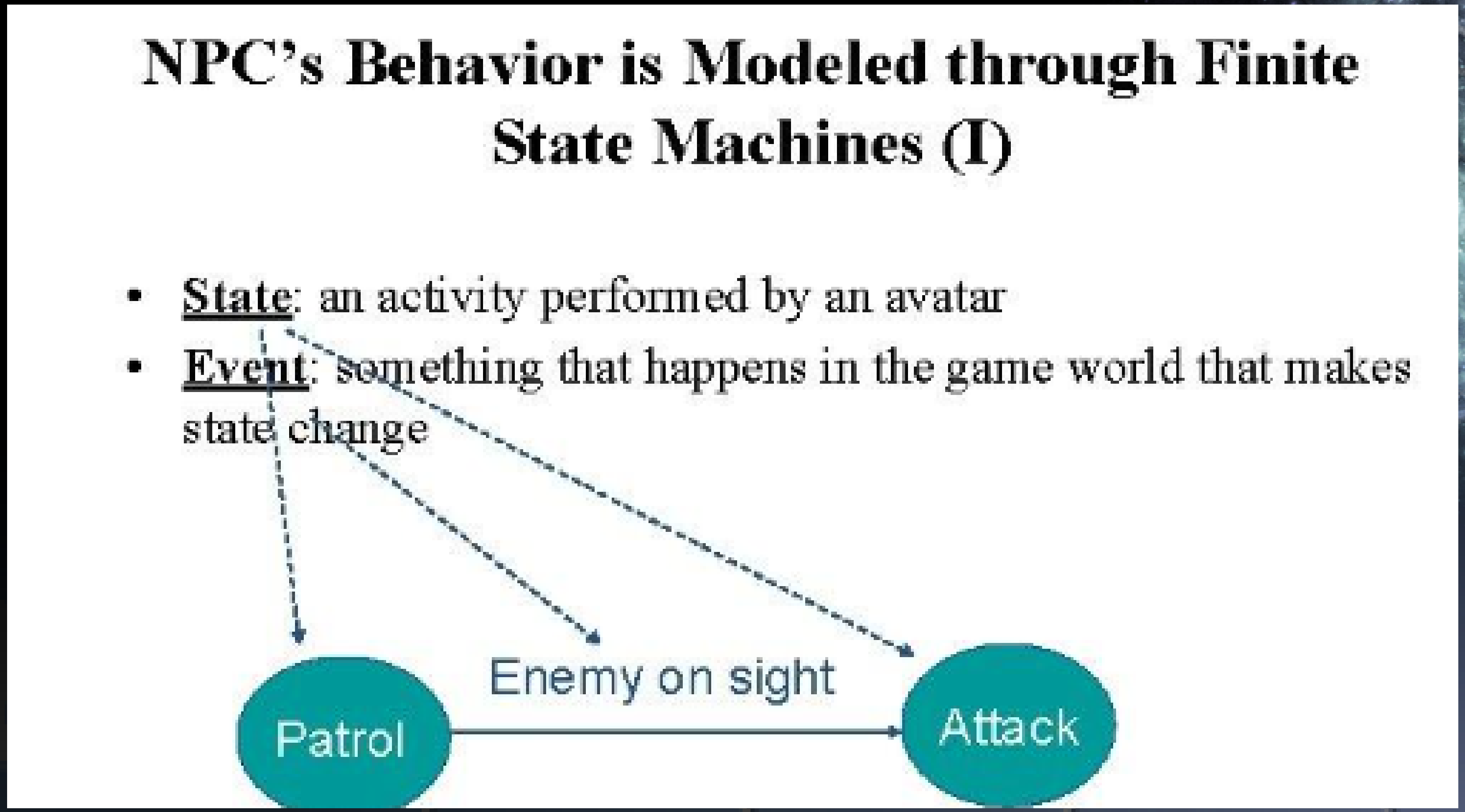
Enemy Nearby?

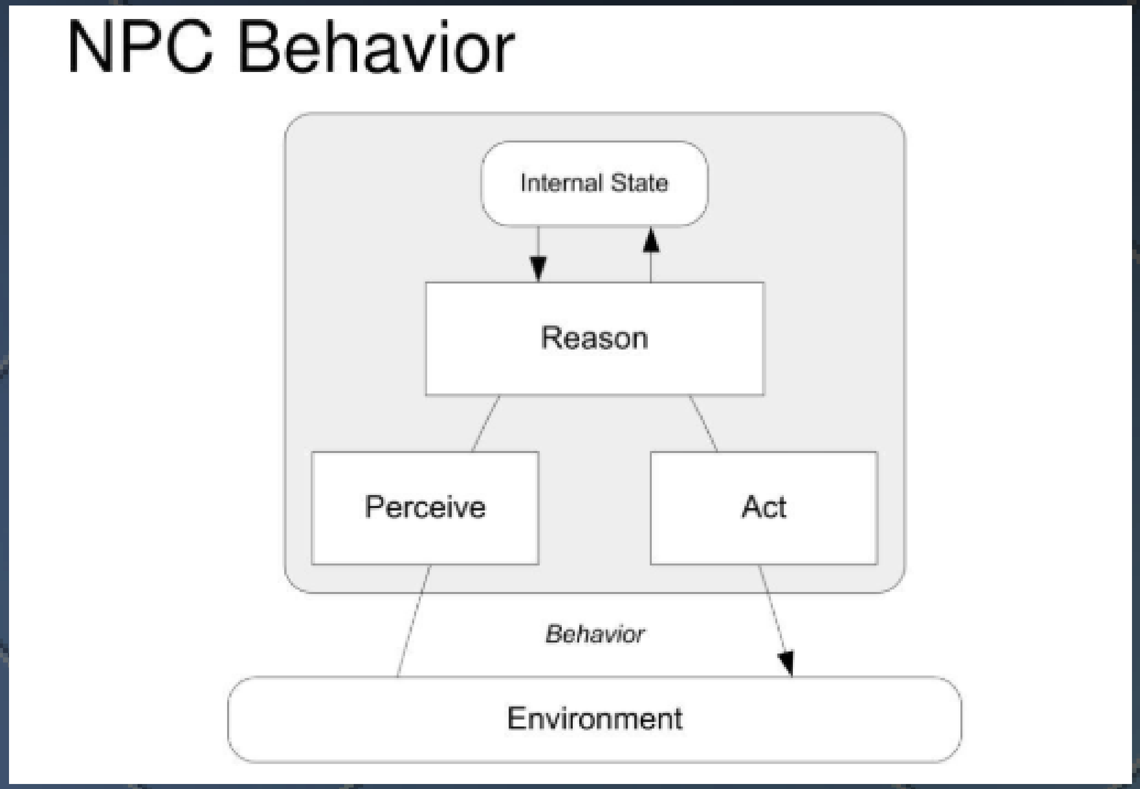
Yes → Attack Enemy

No → Patrol Area




Decision Tree
Is enemy visible?
No
Yes
Is enemy audible?
Is enemy <10 m away?
Yes
No
Yes
Is enemy on flank?
Creep
Attack
No
Yes
Attack
Move


NPC's Behavior is Modeled through Finite State Machines (I)
• State: an activity performed by an avatar
• Event: something that happens in the game world that makes state change
Enemy on sight
Patrol
Attack


NPC Behavior
Internal State
Reason
Perceive
Act
Behavior
Environment

CRUD Operations

- Create → Add Node
- Read → Display Tree
- Update → Modify Decision
- Delete → Remove Node

Algorithm (Insert)

1. Start
2. Check if tree empty
3. Create node
4. Compare IDs
5. Insert left or right

Algorithm (Search)

1. Compare ID with root
2. If equal → Found
3. If smaller → Left
4. If larger → Right

Algorithm (Update & Delete)

Update:

- Search node
- Modify decision

Delete:

- Find node
- Remove it
- Adjust tree

Implementation Details

C Programming

Structures

malloc & free

Functions

Tree traversal

Menu Operations

Add Node

Delete Node

Update Node

Search Node

Display Tree

Exit

Sample Output

===== VIDEO GAME AI =====

1. Add Node

2. Display Tree

Node ID: 10

Decision: Attack Enemy

Advantages

Simple decision making

Easy to modify

Efficient structure

Real-world application

Limitations

- Console-based only
- No graphical view
- Basic AI logic

Future Enhancements

- Graphical tree visualization
- Game engine integration
- Advanced AI logic
- Multiple conditions

Conclusion

- Implemented Decision Tree
- CRUD operations successful
- Useful for Game AI
- Learned data structures & memory

Thank you